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Home.md

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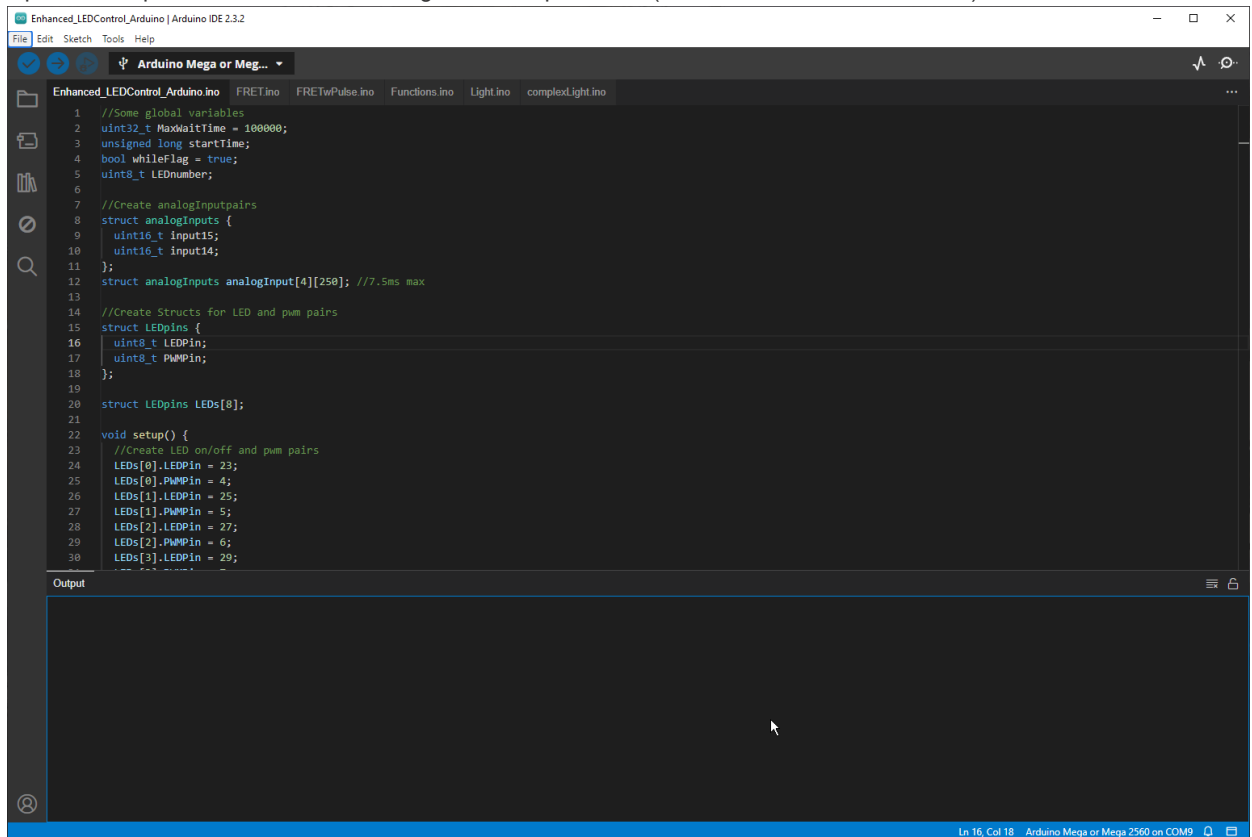
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Arduino.md

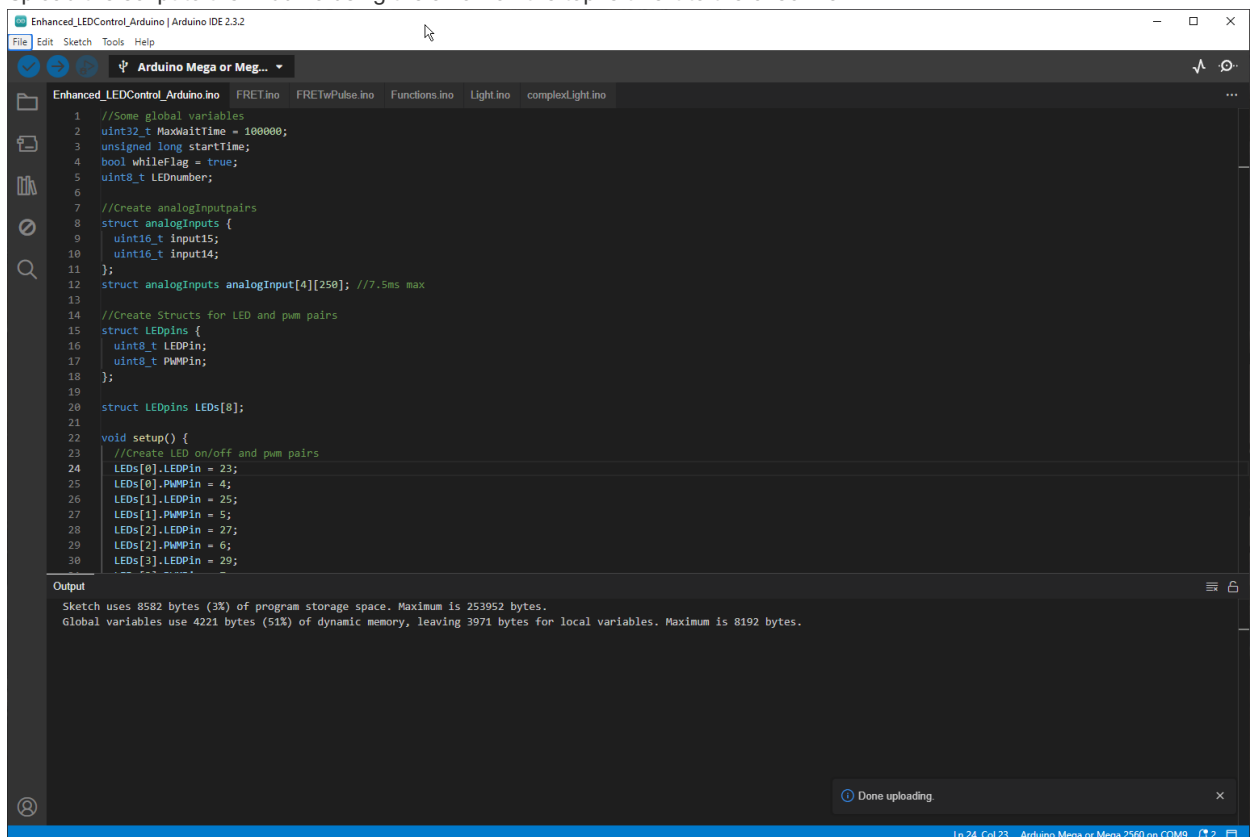
Programming

Prior to use the Arduino unit needs to be programmed:

1. Open the script with the Arduino IDE using the desktop shortcut (Enhanced LEDControl Arduino)



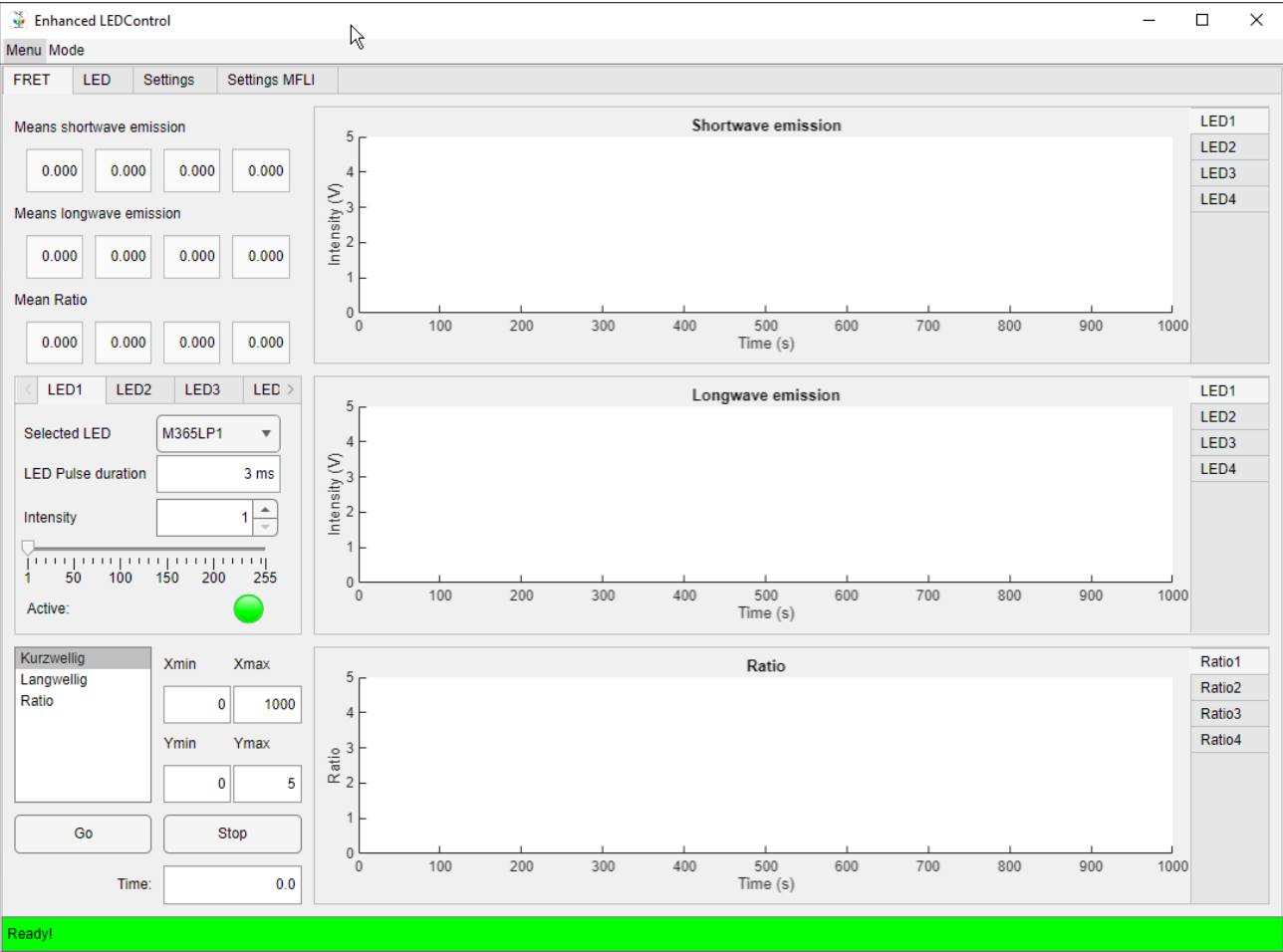
2. Select the Arduino Mega on the right COM Port in top left selection menu.
3. Upload the script to the Arduino using the arrow on the top left next to the checkmark.



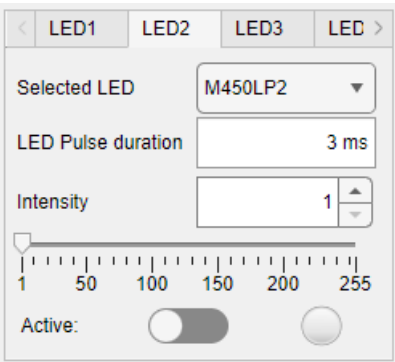
4. On the bottom left a small pop-up window will appear to confirm the upload.

FRET.md

General



- Use the LED tabs on the left to setup the LEDs for the measurement.
- Use the Axis box on the bottom left to setup and change the Plot Axis.
- Use the Plot tabs on the right to switch between plots if more than one LED has been selected.



- A pulse duration of atleast 3 ms is recommended due to the turn on lag of the photodiodes. When changing the illumination time change the marker settings accordingly.
- Use the active switch to select multiple LEDs, which will be illuminated in sequence.

<	LED3	LED4	General	Li	>
Time Pause					
47 ms					
Frequency					
20 Hz					

- Use the "Time Pause" or "Frequency" fields to change the pause between two illumination steps and the measurement frequency.

FRET MFLI

<	LED4	General	Lightpulse	>
Time Pause				
900 μ s				
Frequency				
980.392 Hz				

- Use the Menubars Mode Submenu to change to MFLI mode. Using this mode "time pause" time should be atleast 5 times higher than the pulse time. Note that there is also a 20 μ s switching delay which is already added to the frequency calculation, but it not shown in the "time pause" field.

FRET with lightpulse

<	LED4	General	Lightpulse	>
Selected LED				
Empty				
LED Pulse duration				
1000 ms				
Intensity				
1				
Pulse after:				
10 s				
Active:				
☒ ☐				

- Using the lightpulse tab a lightpulse can be emitted after a fixed time.

Installation.md

Requirements

Hardware

About 20 MB of free space.

Software

[Zurich Instruments Lab One](#) needs to be installed to use the MFLI mode.

Matlab R2024 or Matlab Runtime 2024 needs to be installed.

Connections

FRET Arduino

The 4 pin socket of the arduino unit must be connected to the 4 pin socket of the voltage divider.

FRET MFLI

The 4 pin socket must be connected to the 4 pin to BNC converter. Output 1 and 3 of the converter must be connected to trigger input 1 and 2 of the MFLI amplifier.

LEDControl.md

Single LED excitation

The screenshot displays the 'Enhanced LEDControl' software window. The 'LED' tab is active, showing settings for four LED slots. LED #1 is highlighted with a red box. The 'Selected LED' dropdown for LED #1 is set to 'M365LP1'. The 'Intensity' slider for LED #1 is set to 1. The 'LED pulse duration' is 4000 ms, and the 'LED pulse pause' is 0 ms. The 'Pause between repeats' is 0 ms, and the '# Repeats' is 0. The 'LED #for Single LED' dropdown is set to 'LED #1', and the 'Start Single LED' button is visible. The 'Complex LED excitation' section on the right shows 'LED #1' selected. The 'Protocol' graph shows a series of pulses between 400 nm and 600 nm over 900 ms. A green bar at the bottom indicates 'Ready!'.

LED Slot	LED #1	LED #2	LED #3	LED #4
Selected LED	M365LP1	M450LP2	M595L4	Empty
Intensity	1	1	1	1
LED pulse duration	4000 ms	4000 ms	1000 ms	1000 ms
LED pulse pause	0 ms	0 ms	0 ms	0 ms
Pause between repeats	0 ms			
# Repeats	0			
LED #for Single LED	LED #1	Start Single LED		

Complex LED excitation:

- LED #1
- LED #2
- None
- None
- ☐ Switch LED on after Complex
- LED #1
- Start Complex
- Stop

Protocol

Wavelength (nm) vs Time (ms) graph showing a series of pulses between 400 nm and 600 nm over 900 ms.

Ready!

1. Select the LED you want to use in the "selected LED" row.
2. Set the wanted intensity in the "Intensity". The intensity can be changed later.
3. Select the wanted LED column in the LED# for Single LED row.
4. Press Start Single LED to start the illumination.

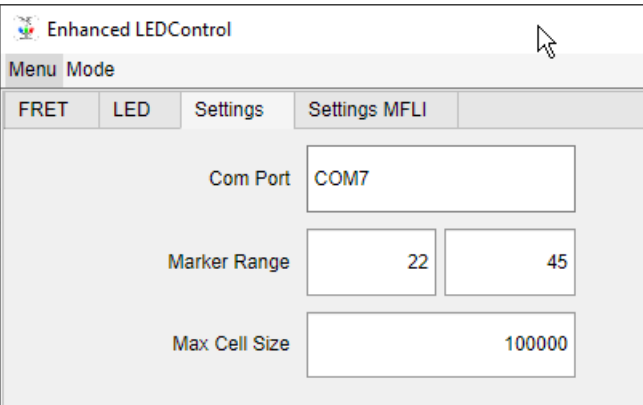
Complex LED excitation



1. Select the amount of LEDs that you want to use and their order in the complex LED excitation column. The LED numbers are from the LED Slot row.
2. Select the LED you want to use in the "selected LED" row.
3. Set the wanted intensity in the "Intensity" row. The intensity can be changed later.
4. Change on and off duration of the LEDs in the "LED pulse duration" and "LED pulse pause" rows.
5. Set the amount of repeats and the pause between each repeat in the "# Repeats" and "Pause between repeats" rows.
6. Control the illumination pattern in the protocol preview plot.
7. If you want to switch a single LED on after the complex protocol mark the "Switch LED on after Complex" box.
8. Start the protocol by clicking the "Start Complex" button.

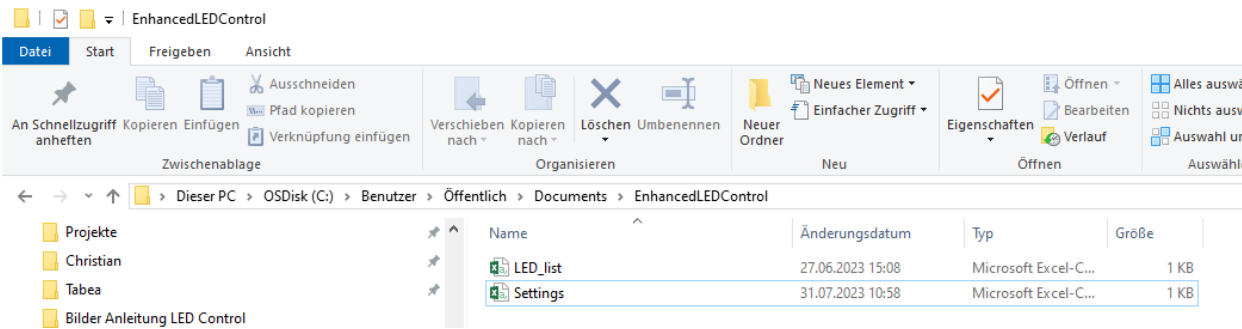
Settings.md

General



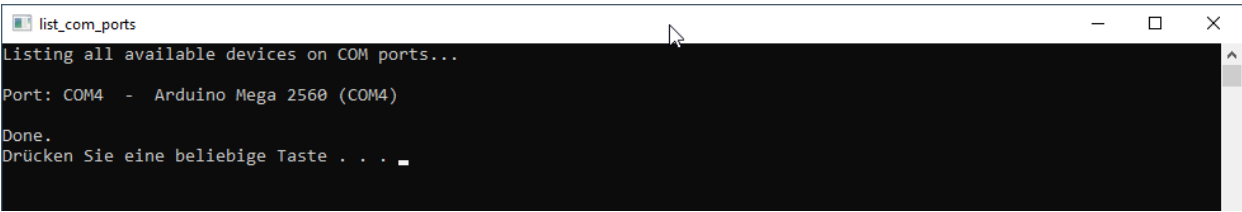
- COM Port: The COM Port is needed to connect to the Arduino. Change to the current one.
- Marker Range: The marker range indicates where mean values are calculated from the analog input signal. Do not change this unless you know what you are doing.
- Max Cell Cize: The size of the preallocated data array. Do not change this unless you know what you are doing.

Storage location of the settings



- Settings and the LED List are saved under: C:\Users\Public\Documents\EnhancedLEDControl

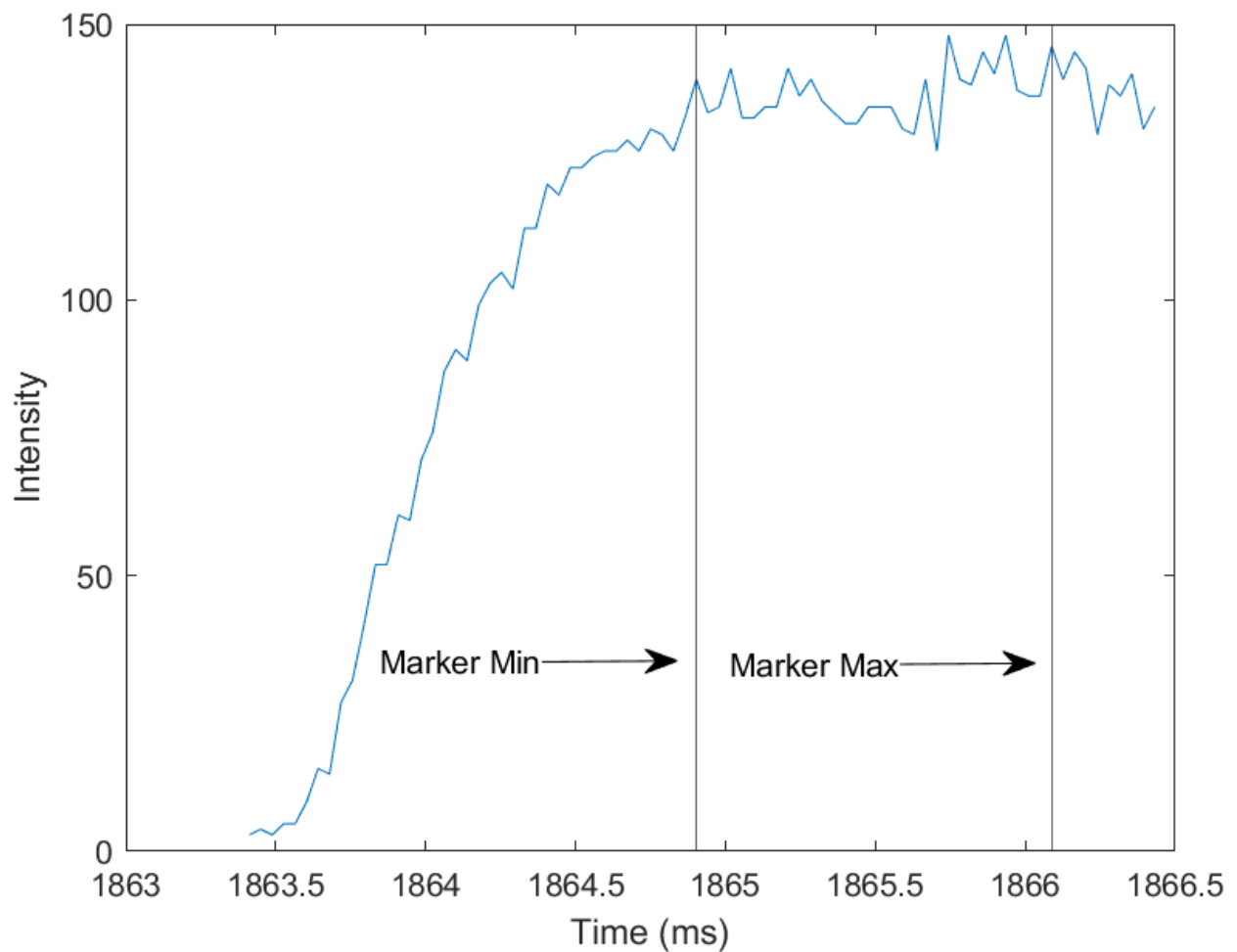
COM Port



- Use the list_com_ports.bat script to check which COM Port is connected to the Arduino unit.

Settings Arduino

Marker Range



- Only the data inbetween the markers is used to calculate the mean, which will be displayed and saved.

LEDs

Enhanced LEDControl

Menu Mode

FRET LED Settings Settings MFLI

Com Port

COM7

Marker Range

22

45

Max Cell Size

100000

	Name	Wavelength [nm]	Filter wavelength [nm]
LED1:	M365LP1	365	F39-363: 346-373 nm
LED2:	M450LP2	440	F37-433: 419-443 nm
LED3:	M595L4	585	F39-572: 555-587 nm
LED4:	Empty	505	F39-508: 497-520 nm
LED5:	Empty	530	F39-833: 521-542 nm
LED6:	Empty	585	F39-572: 555-587 nm
LED7:	Empty	630	F39-641: 627-654 nm
LED8:	Empty	405	F37-406: 396-417 nm

- The LEDs, their peak wavelengths and filters are stored in the LEDlist.csv. The names are purely cosmetic and may not be the LEDs connected to the Arduino unit. Make sure the names and the LEDs are identical.

MFLI

Enhanced LEDControl

Menu Mode

FRET LED Settings Settings MFLI

IP Address 10.173.128.48

Device ID dev5094

Port 8004

Api Level 6

Check Connection

Off On

Low Data Save!

- IP Address: The IP address is needed to connect to the MFLI lock-in amplifier. Sometimes the IP changes due to network changes. The current IP can be viewed in the LabOne software.

Config Device Aux Lo

Web Server About Docs

Version	24.04.57831
Host	127.0.0.1
Port	8006

Data Server

Version	24.04.57831
Host	10.173.128.48
Port	8004

Connectivity From Everywhere ▼

File Upload

Drop settings file(s) here

- Device ID: The serial number of the amplifier. Do not change this unless you are changing the amplifier.
- Port: Port of the amplifier. Do not change this value. The current port can be seen in the LabOne software.
- API Level: The API level required to communicate with the amplifier. Do not change.
- Check Connection: Test if FRETControl is able to connect to the MFLI Lock-In amplifier with the current settings.